

RIFAT ERDEM SAHIN

Machine Learning Lead | AI Strategy & Engineering Specialist

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Schedule a Call: <https://calendly.com/rifaterdem/schedule>

PROFESSIONAL SUMMARY

Senior Machine Learning Lead specializing in **AI Strategy, Large-Scale ML Systems, and Generative AI (LLMs)**. Expert in leading cross-functional engineering teams to design, build, and deploy production-grade ML models. Proven track record in delivering AI solutions that achieved 300% productivity improvements and 40% reduction in operational overhead across financial services, healthcare, and global technology sectors. Expert in MLOps, model governance, and AI-powered automation using IBM watsonx and cloud-native ML platforms.

CORE COMPETENCIES

AI/ML Leadership & Strategy - Leading ML engineering teams and AI research initiatives - AI product roadmap definition and architectural oversight - Generative AI & LLM strategy (Fine-tuning, RAG, Prompt Engineering) - Model Governance, Ethics, and Responsible AI frameworks

ML Engineering & MLOps - Production-grade ML pipelines (Kubeflow, MLflow, Airflow) - Scalable model serving (Kubernetes, Triton, BentoML) - Feature Store design and data engineering for ML - Continuous Integration and Deployment for ML (CI/CD for ML)

Data Science & Research - Advanced Statistical Modeling and Deep Learning (PyTorch, TensorFlow) - NLP, Computer Vision, and Time-Series Forecasting - Experiment tracking and model performance optimization - Data-driven decision making and stakeholder management

KEY ACCOMPLISHMENTS

2025 | IBM | London, UK

Enterprise Generative AI & ML Platform Leadership - Challenge:

Lead the development of enterprise-scale Generative AI solutions and automate complex ML workflows for global clients - **Solution:** Architected a comprehensive AI platform leveraging IBM watsonx.ai for rapid model development and automated deployment on Red Hat OpenShift - **Impact:** - 35% reduction in model development cycle time via automated MLOps pipelines - 40% improvement in AI-driven insight accuracy using custom-tuned LLMs - Successfully scaled AI solutions to support mission-critical operations across 5+ global accounts - **Technologies:** IBM watsonx.ai, Red Hat OpenShift, PyTorch, LangChain, Hugging Face, Terraform, GitHub Actions

2024 | Goldman Sachs | Muscat, Oman

Enterprise ML Financial Risk Platform - Challenge: Design and lead the implementation of a high-performance ML platform for real-time financial risk assessment - **Solution:** Architected a hybrid-cloud ML platform with automated model retraining and real-time inference capabilities - **Impact:** - 300% improvement in risk detection speed through optimized ML architecture - 30% reduction in operational costs via intelligent resource management for ML workloads - 99.9% availability for mission-critical ML-driven financial workflows - **Technologies:** Kubernetes, AWS SageMaker, MLflow, Spark, Python, Terraform

2023 | Ypsomed | Switzerland

Healthcare IoT Predictive Analytics - Challenge: Lead the development of predictive maintenance models for medical IoT devices - **Solution:** Developed an end-to-end ML solution for real-time device health monitoring and anomaly detection - **Impact:** - 40% improvement in device uptime through predictive maintenance - 35% reduction in field service costs via AI-driven remote diagnostics - Real-time patient safety insights enabling proactive healthcare delivery - **Technologies:** Azure ML, IoT Hub, Python, Scikit-learn, Docker, Grafana

TECHNICAL EXPERTISE

AI & Machine Learning

Frameworks: PyTorch, TensorFlow, Scikit-learn, JAX, XGBoost

Generative AI: LLMs (GPT, Llama, Granite), LangChain, RAG, Vector DBs (Milvus, Pinecone)

NLP & CV: Transformers, BERT, OpenCV, Image Segmentation
Tools: Jupyter, Weights & Biases, MLflow, DVC

ML Infrastructure & DevOps

Platforms: Kubernetes, Red Hat OpenShift, AWS SageMaker, GCP Vertex AI, Azure ML
MLOps: Kubeflow, Airflow, ZenML, Feast (Feature Store)
Automation: Terraform, Ansible, GitHub Actions, GitLab CI/CD
Monitoring: Prometheus, Grafana, ELK Stack, Evidently AI

PROFESSIONAL EXPERIENCE HIGHLIGHTS

Senior Machine Learning Lead / AI Solutions Architect | January 2025
- Present
IBM / London, UK

- Leading the delivery of **Generative AI solutions** using **IBM watsonx.ai**, enabling enterprise clients to leverage LLMs for automated business processes
- Architecting **MLOps frameworks** on Red Hat OpenShift, standardizing model deployment and monitoring across global teams
- Driving the adoption of **Responsible AI** practices, ensuring model fairness, transparency, and explainability
- Mentoring a team of 15+ ML engineers and data scientists on advanced AI techniques and production engineering
- Collaborating with C-suite stakeholders to define AI transformation strategies and identify high-impact use cases

Senior ML Lead / Technical Lead | 2020 - 2025
Goldman Sachs, Ypsomed, Cushman & Wakefield

- Led ML engineering initiatives across finance, healthcare, and real estate sectors
- Designed and implemented **End-to-End ML systems** supporting millions of daily transactions
- Established organizational ML best practices and automated validation frameworks
- Mentored engineering teams on modern ML technologies and MLOps practices
- Delivered solutions achieving 300% improvement in operational efficiency through AI

Machine Learning Engineer | 2016 - 2020
Microsoft, Emerson, Various Fortune 500

- Built enterprise ML solutions for digital transformation initiatives

- Led cross-functional teams designing scalable AI applications
 - Established model deployment frameworks and implementation strategies
 - Evangelized AI technologies through technical leadership and workshops
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EDUCATION

Bachelor of Science
Southern New Hampshire University, USA | 2013

CERTIFICATIONS & CONTINUOUS LEARNING

AWS Certified Machine Learning - Specialty
Google Cloud Professional Machine Learning Engineer
Certified Kubernetes Administrator (CKA)
DeepLearning.AI Specializations (NLP, GANs, etc.)

SECURITY CLEARANCES

UK SC (Security Check) - Valid until 2028
NATO Clearance - Valid until 2029
Background Checks: Watchdog (2024), Sterling (2019)

AVAILABILITY & CONTACT

Immediate Availability for Machine Learning Lead roles
Schedule a Discussion: <https://calendly.com/rifaterdem/schedule>
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Phone: +44 7848 024173

SUPPORTING DOCUMENTS

Download Full CV (PDF):
https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.pdf

Download Full CV (Word):
https://rifaterdemsahin.com/wp-content/uploads/2025/05/erdem-sahin-cv_summary_2025_may.docx

Technical Portfolio & Presentations:

<https://rifaterdemsahin.com/wp-content/uploads/2025/02/rifaterdemsahinprofilepresentation.v2025.2.pdf>

References and detailed project portfolios available upon request